

TASK 5: EXPLORATORY DATA ANALYSIS (EDA) ON TITANIC DATASET

1. Objective

The objective of this project is to perform Exploratory Data Analysis (EDA) on the Titanic dataset to identify patterns, trends, and relationships among passenger attributes. Statistical summaries and visualizations are used to gain insights into factors that influenced passenger survival.

2. Dataset Description

The Titanic dataset contains information about passengers aboard the RMS Titanic. The dataset includes demographic information, passenger class, age, gender, ticket fare, cabin information, embarkation point, and survival status.

The dataset is widely used for data analysis and machine learning projects because it contains both numerical and categorical variables and provides a real-world classification problem.

3. Data Exploration

The dataset was explored using various Pandas functions including:

- `df.info()`
- `df.describe()`
- `df.isnull().sum()`
- `value_counts()`

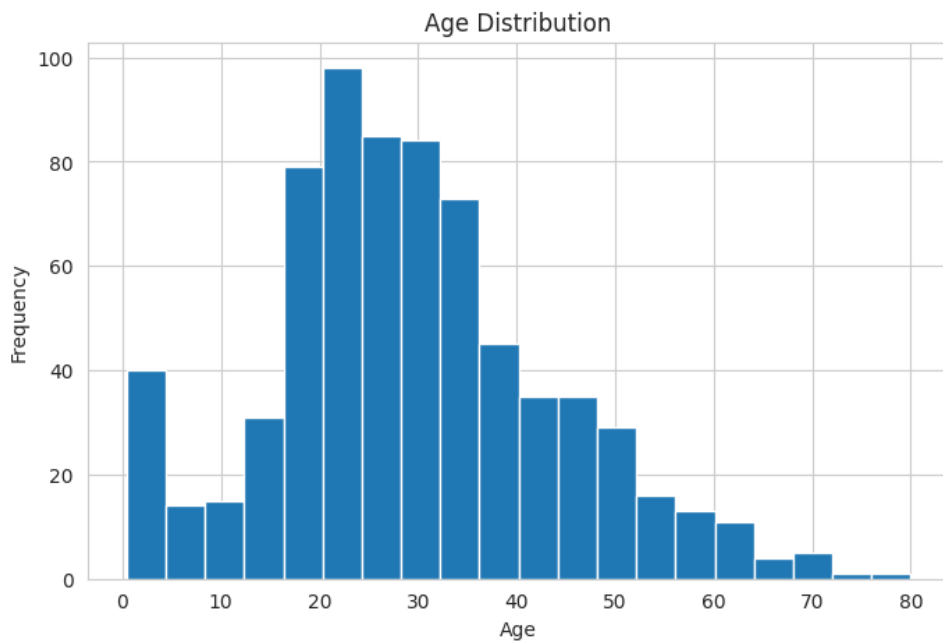
These functions helped identify data types, missing values, statistical summaries, and class distributions.

Findings from Initial Exploration

- The dataset contains 891 passenger records.
 - Multiple numerical and categorical attributes are present.
 - Missing values exist in Age, Cabin, and Embarked columns.
 - Survival information is available for all passengers.
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4. Visualizations and Observations

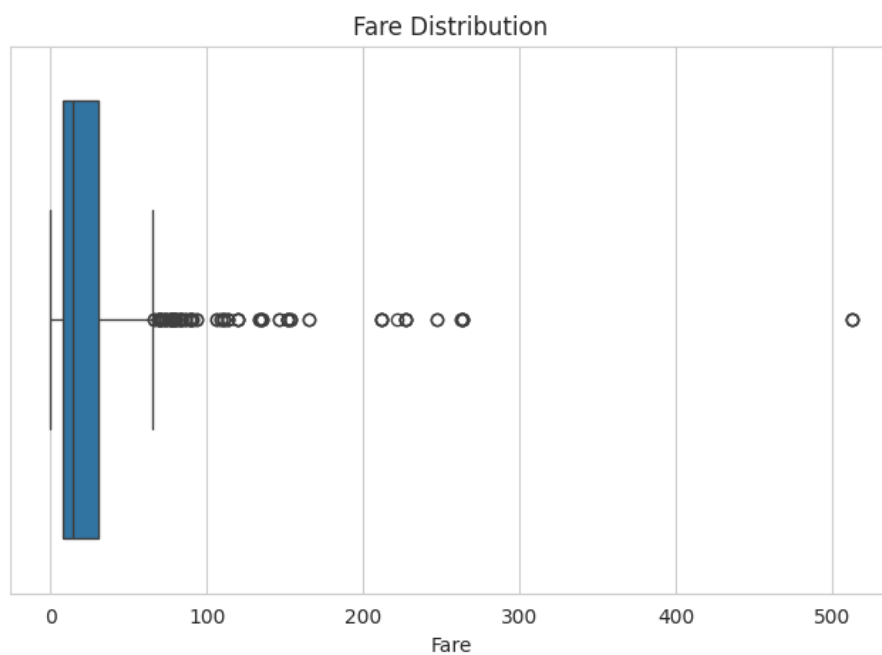
Graph 1: Age Distribution Histogram



Observation

Most passengers were between 20 and 40 years old. The dataset contains fewer elderly passengers and young children compared to adults.

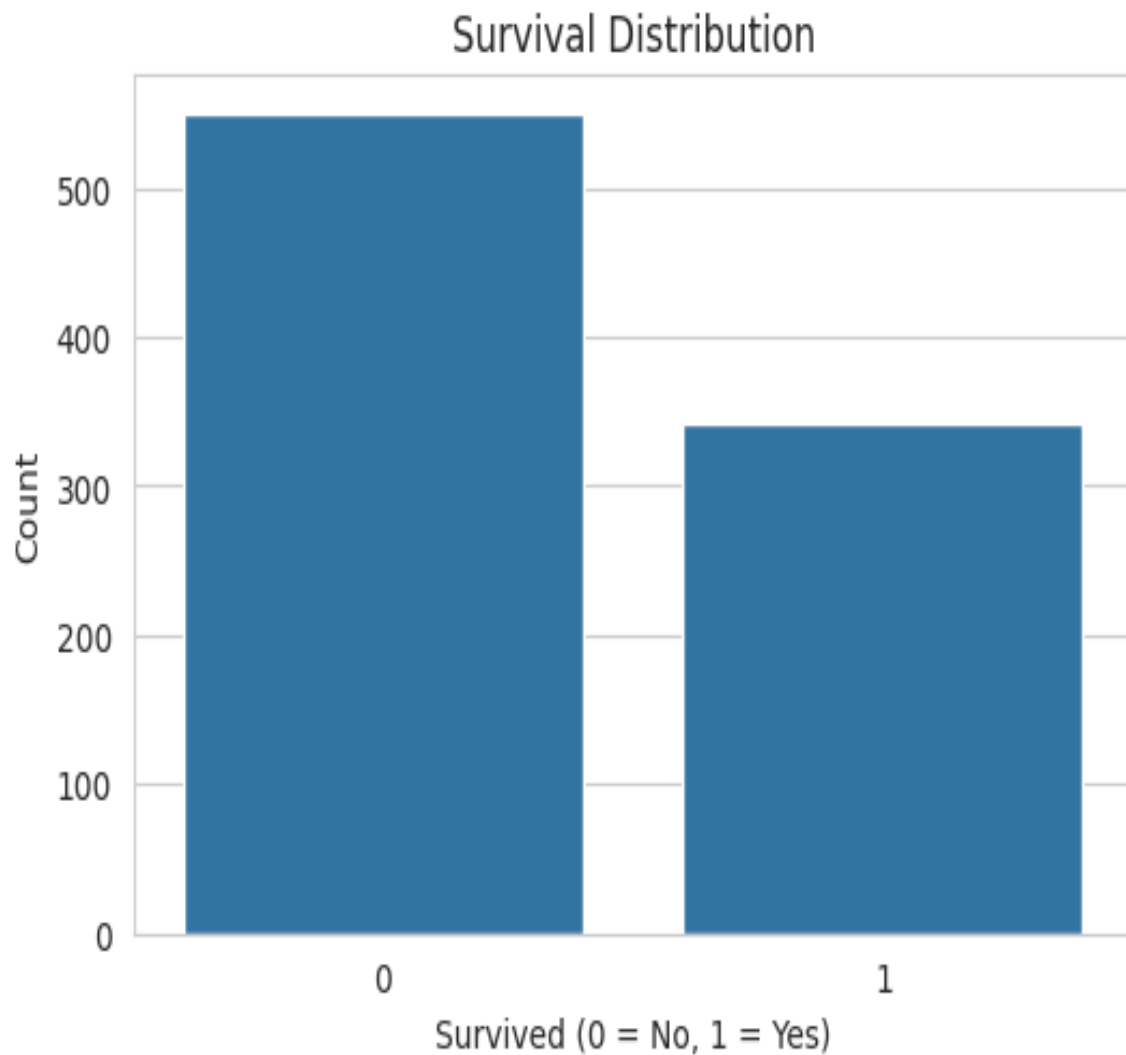
Graph 2: Fare Distribution Boxplot



Observation

The Fare column contains several outliers, indicating that a small number of passengers paid significantly higher ticket prices than the majority.

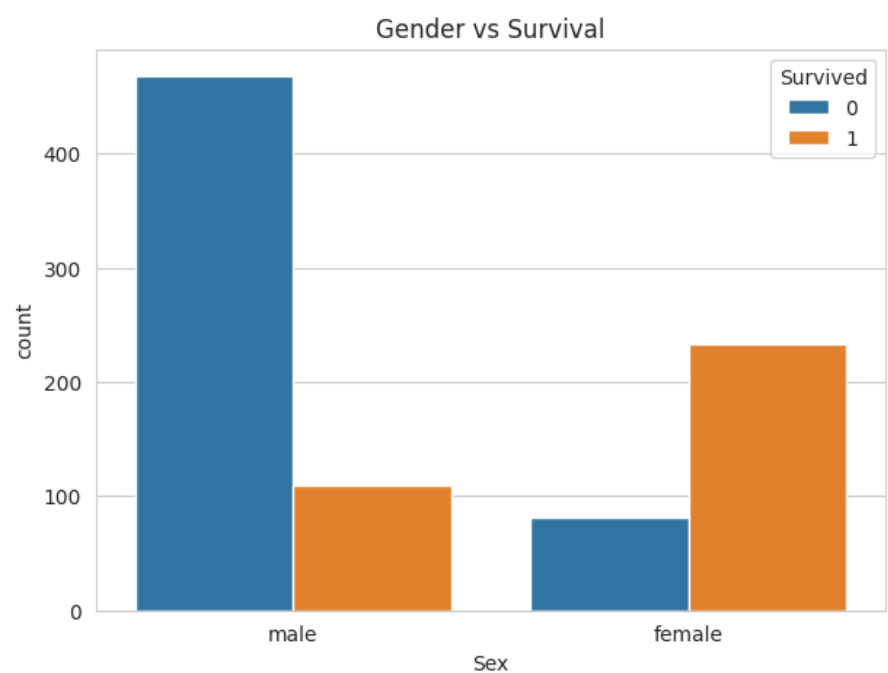
Graph 3: Survival Distribution



Observation

The number of passengers who did not survive was greater than the number of passengers who survived.

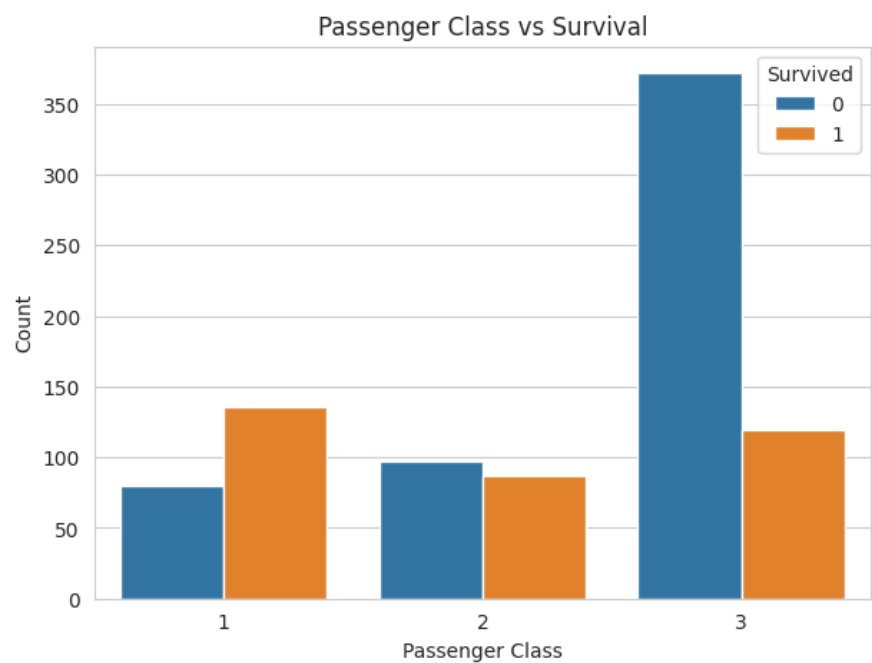
Graph 4: Gender vs Survival



Observation

Female passengers had a significantly higher survival rate compared to male passengers, indicating that gender played an important role in survival outcomes.

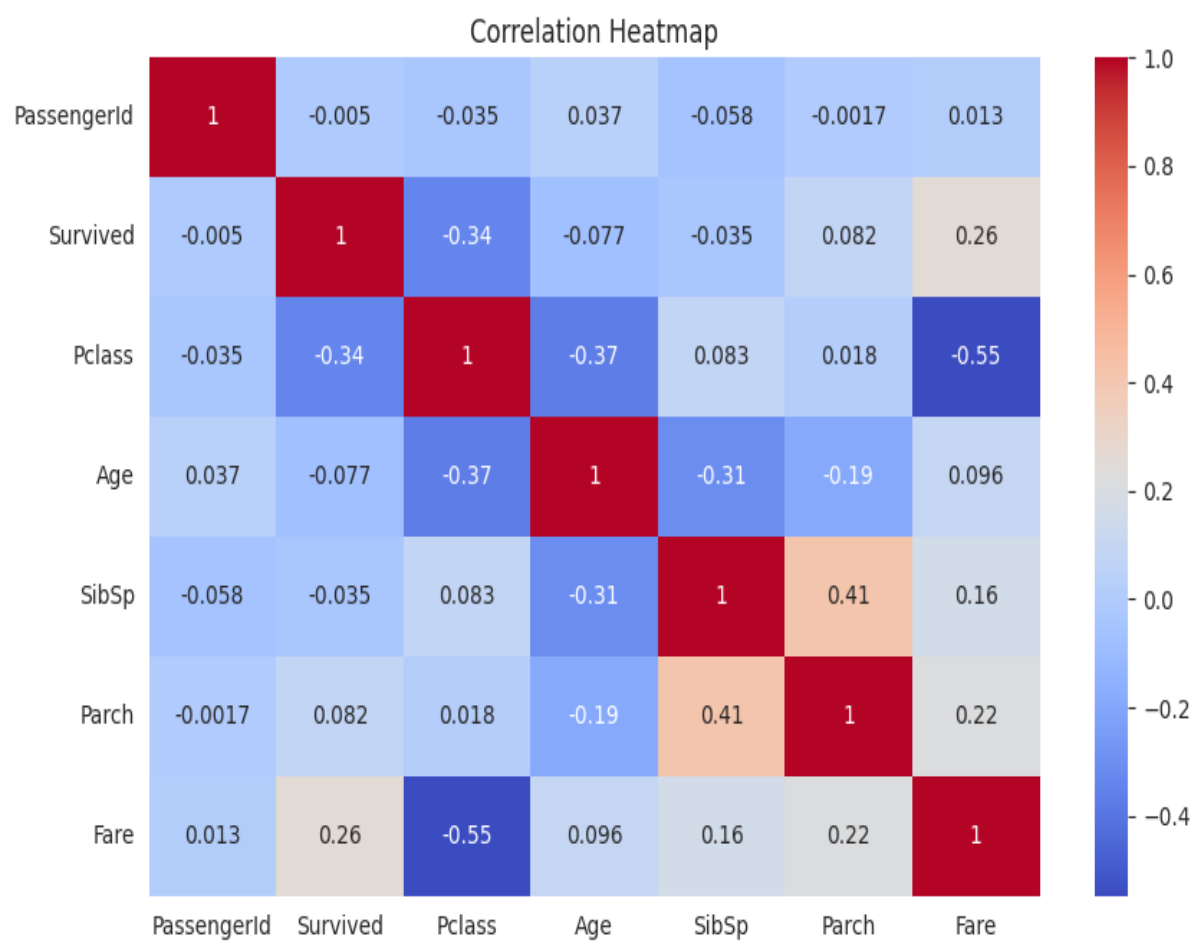
Graph 5: Passenger Class vs Survival



Observation

Passengers traveling in first class had a higher survival rate compared to passengers in second and third class.

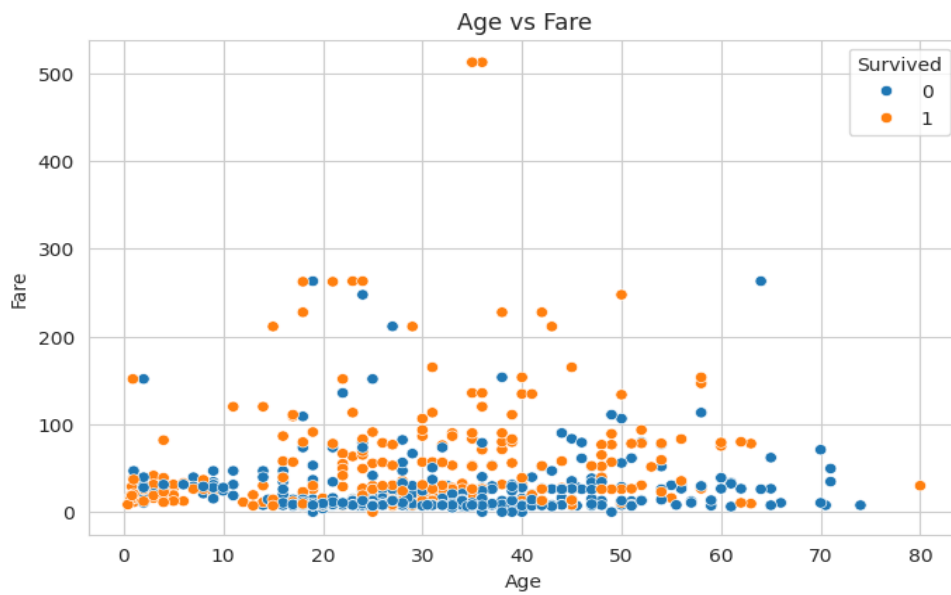
Graph 6: Correlation Heatmap



Observation

The heatmap shows relationships among numerical variables. Fare exhibits a positive relationship with survival, while passenger class shows a negative relationship with survival.

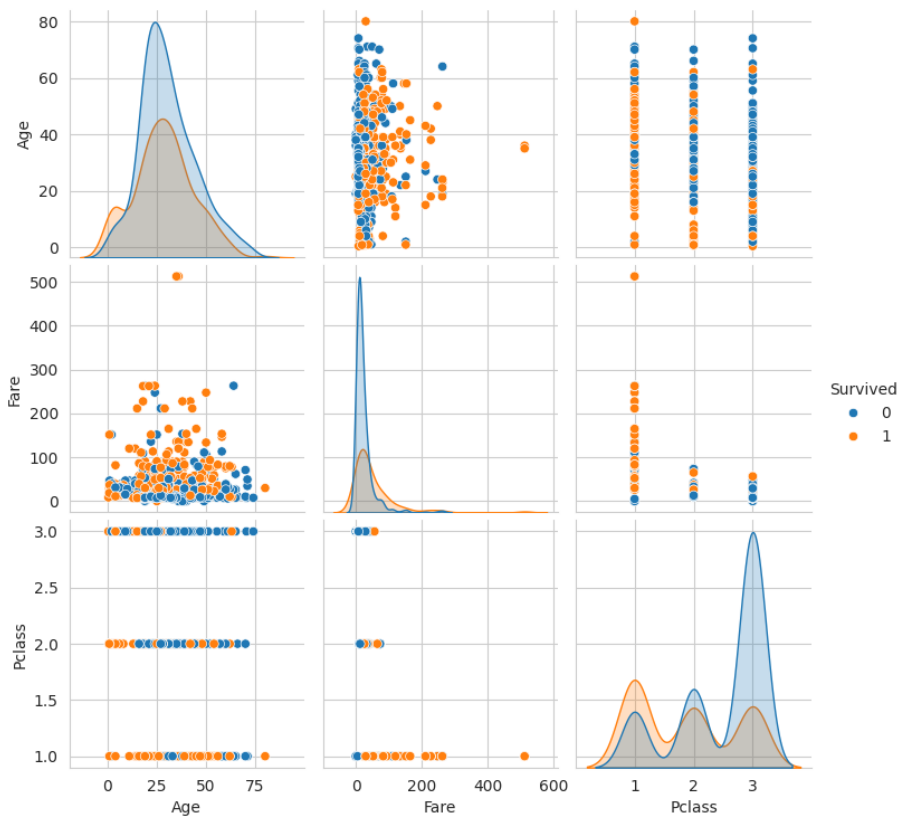
Graph 7: Age vs Fare Scatter Plot



Observation

Passengers who paid higher fares generally showed better survival outcomes. No strong relationship is observed between age and fare.

Graph 8: Pairplot



Observation

The pairplot highlights relationships among Age, Fare, Passenger Class, and Survival. Survival appears more common among passengers with higher fares and better passenger classes.

5. Summary of Findings

The Exploratory Data Analysis performed on the Titanic dataset revealed several important insights regarding passenger survival.

Key findings include:

- Most passengers were adults between 20 and 40 years of age.
- Female passengers had significantly higher survival rates than male passengers.
- First-class passengers were more likely to survive compared to passengers in lower classes.
- Ticket fare showed a positive relationship with survival.
- Several outliers were identified in the Fare column.
- Passenger class, gender, and fare were among the most influential factors affecting survival.

Overall, the analysis demonstrates how data visualization and statistical exploration can uncover meaningful patterns and trends within a dataset.

6. Conclusion

The Titanic dataset was successfully analyzed using Exploratory Data Analysis techniques. Various statistical summaries and visualizations were created to understand the structure of the dataset and identify important factors influencing passenger survival. The project helped develop skills in data cleaning, visualization, pattern recognition, and interpretation of analytical results.